



VMWARE VSPHERE 7.0: INSTALL, CONFIGURE, MANAGE



Lab 02: Installing and Configuring an ESXi Host

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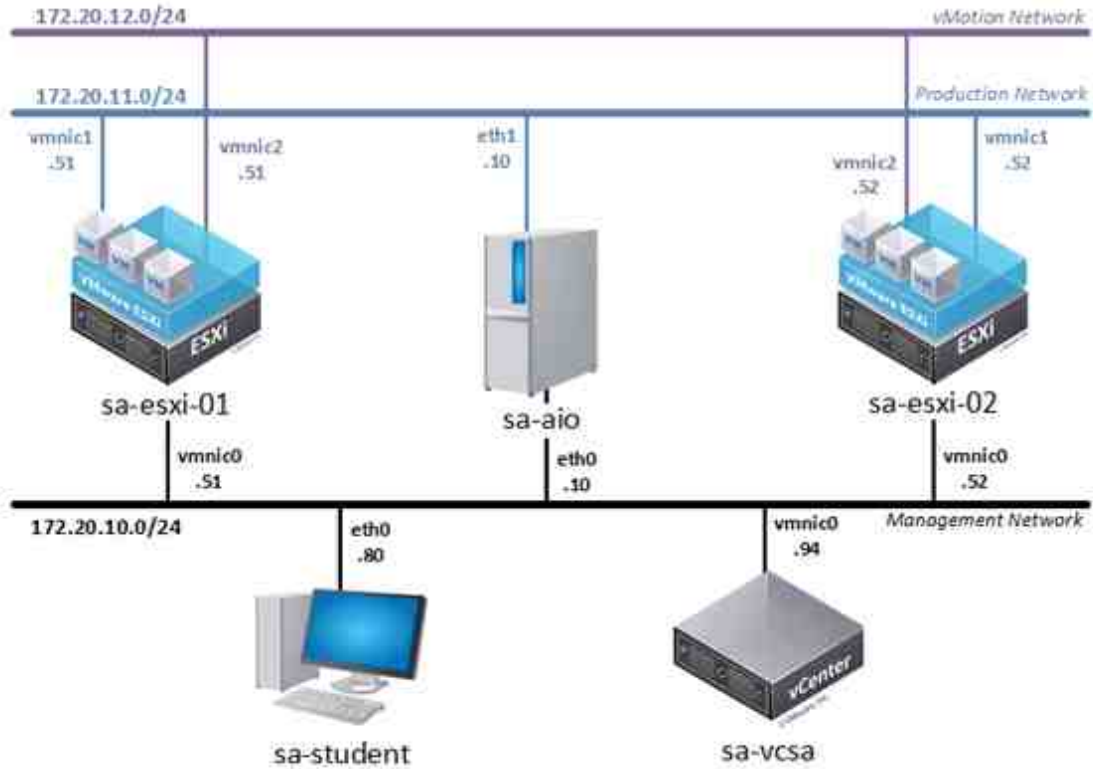
Introduction

In this lab, you will install and configure the VMware ESXi operating system using your student workstation.

Objectives

-) Install ESXi
-) Configure ESXi
-) Enable ESXi Shell and SSH
-) Add an ESXi host to Active Directory
-) Log in to the ESXi host as an Active Directory User

Lab Topology



Lab Settings

The information in the table below will be needed to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
sa-esxi-01	eth0: 172.20.10.51	root	Train1ng\$
sa-esxi-02	eth0: 172.20.10.52	root	Train1ng\$
sa-aio	eth0: 172.20.10.10	sysadmin	vmware123
sa-student	eth0: 172.20.10.80	sysadmin	vmware123
sa-vcasa	eth0: 172.20.10.94	sysadmin@vclass.local	vmware123

1 Installing ESXi

In this task, you will access and manage the lab environment from the student workstation. This system will serve as an end-user terminal with direct access for the VMware infrastructure. You will install VMware ESXi on a virtual machine. This “nested” ESXi host operates in the same way as a physical host but uses different user interfaces for the installation. You will use the vSphere Client to configure and manage the nested ESXi host.

1. Launch the sa-student virtual machine to access the graphical login screen.

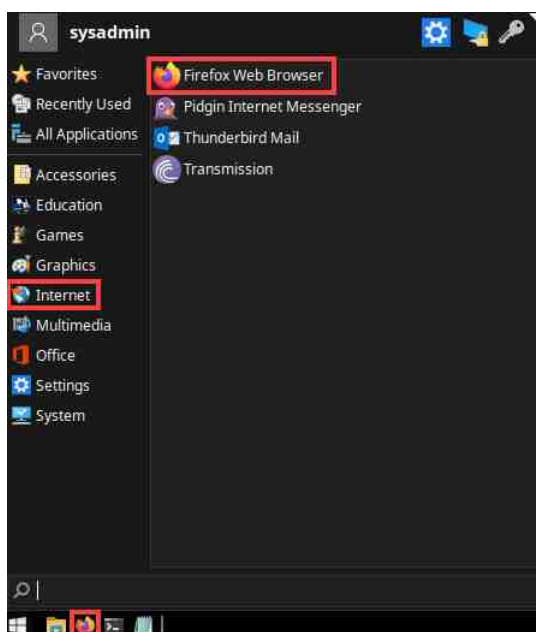


To launch the console window for a virtual machine, you may access by either clicking on the machine's graphic image from the topology page or by clicking on the machine's respective tab from the navigation bar.

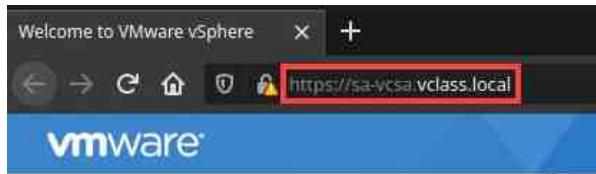
2. Log in as `sysadmin` using the password `vmware123`.



3. Launch the Mozilla Firefox web browser by either clicking on the icon shortcut found on the bottom toolbar or by navigating to Start Menu > Internet > Firefox Web Browser.



4. Notice the homepage is automatically configured to load the URL address of the VMware vCenter Server Appliance (sa-vcasa virtual machine). Click on the LAUNCH VSPHERE CLIENT (HTML5) button to launch the vSphere Client.



Getting Started

LAUNCH VSPHERE CLIENT (HTML5)

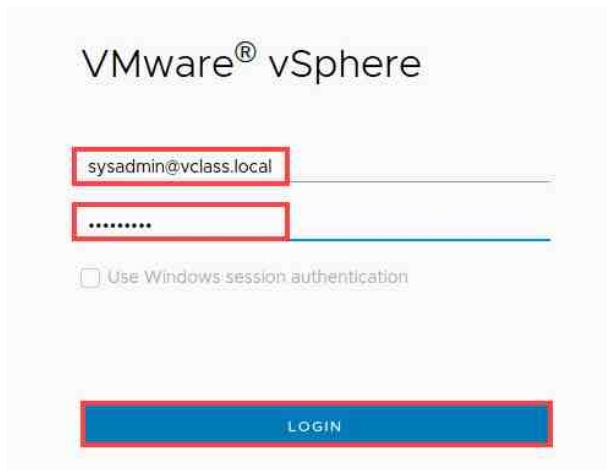
Documentation

[VMware vSphere Documentation Center](#)



If you cannot successfully load the VMware Getting Started webpage, then please wait an additional 3-5 minutes and refresh the page to continue. During this time, the vCenter Server Appliance is still booting up and requires extra time to initialize.

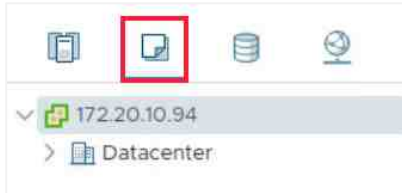
5. In the Username text field, type `sysadmin@vclass.local` and in the Password text field, type `vmware123`. Click on LOGIN.



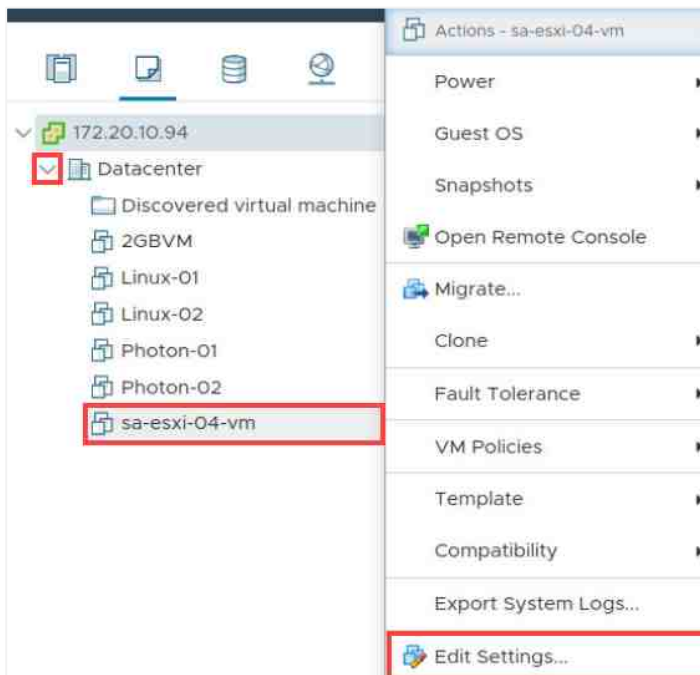
Please Note

You may ignore the “browser-OS combination” warning message presented on the VMware vCenter Single Sign-On page and continue moving forward with the lab.

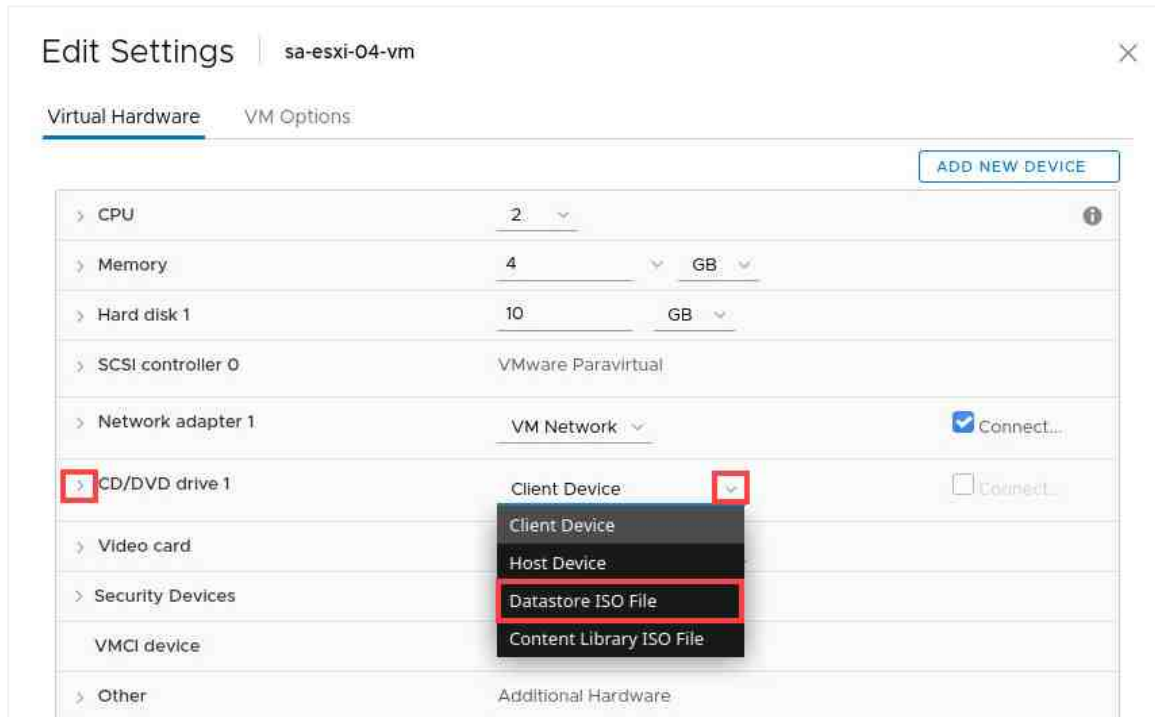
6. While logged in to the vSphere Client, in the navigator pane, select the VMs & Templates tab.



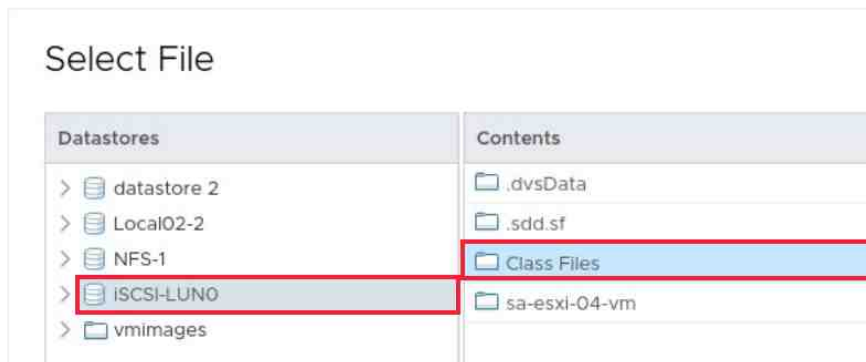
7. Expand the Datacenter object in the navigator pane and right-click sa-esxi-04-vm, followed by selecting Edit Settings.



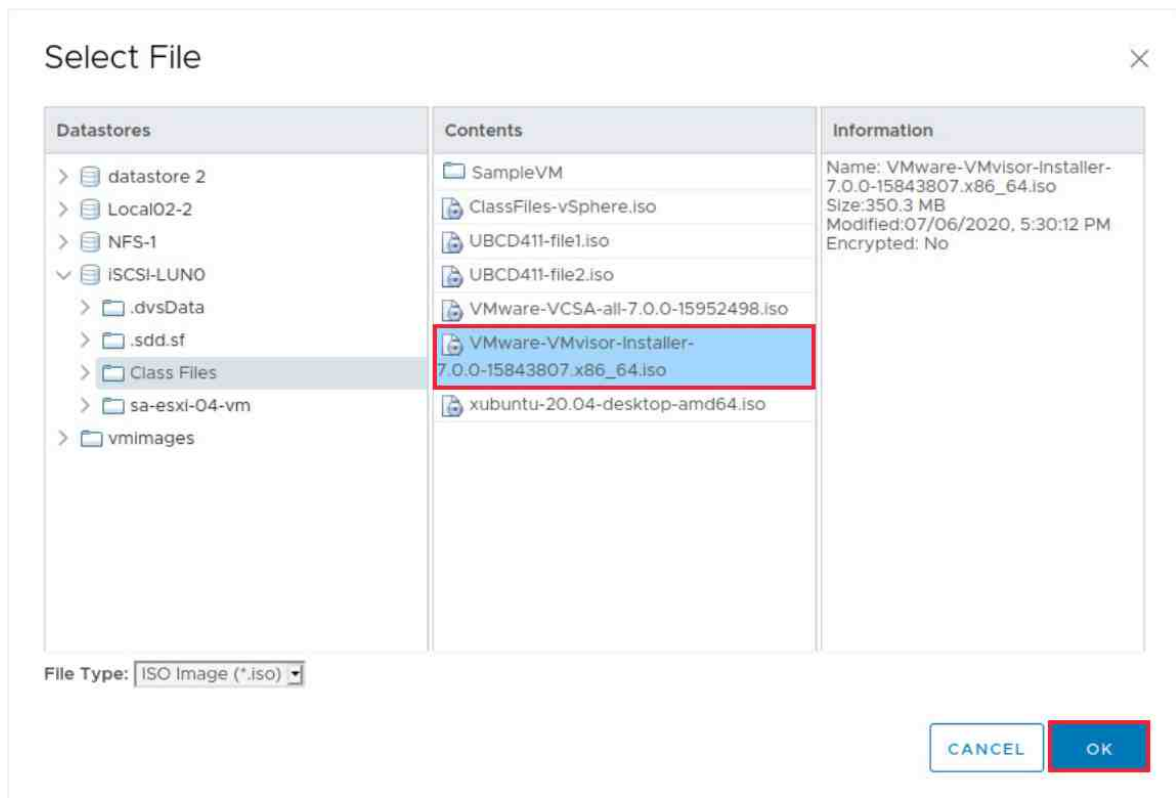
8. In the Edit Settings window, locate the CD/DVD drive 1 entry and click the dropdown button next to Host Device. From the dropdown menu, select Datastore ISO File.



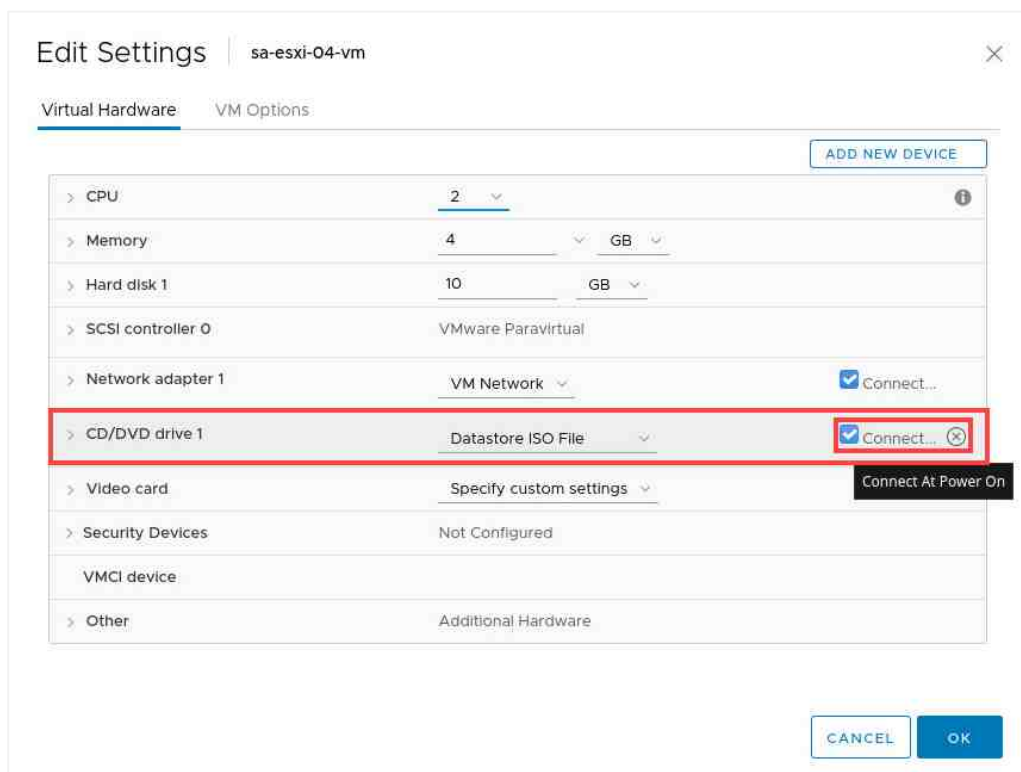
9. In the Select File window, select iSCSI-LUN0 underneath the Datastores pane, followed by clicking the Class Files folder in the Contents pane.



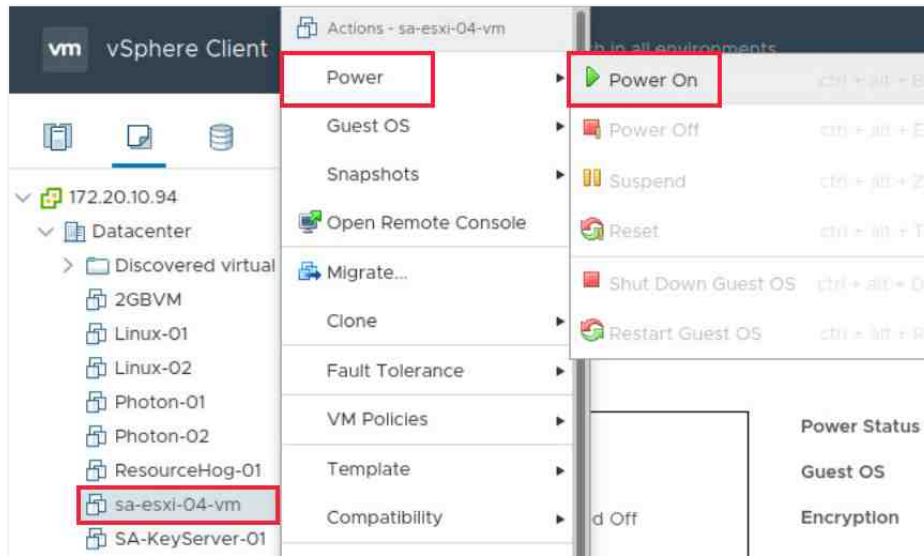
10. Once the Class Files folder is selected, notice the files available. Select the VMware-VMvisor-Installer-7.0.0-15843807.x86_64.iso file. Click OK.



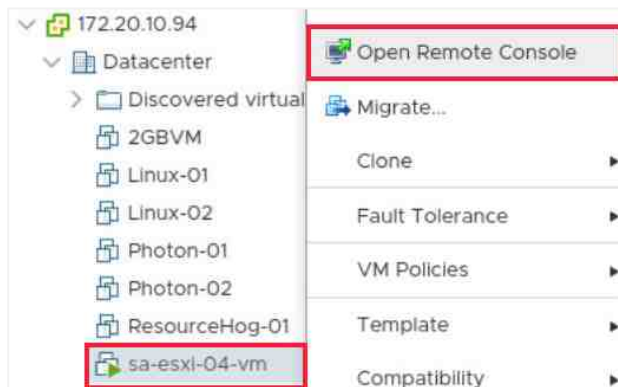
11. Back in the Edit Settings window, select CD/DVD drive 1. Ensure that Connect... (Connect At Power On) is checked. Click OK to save the configurations.



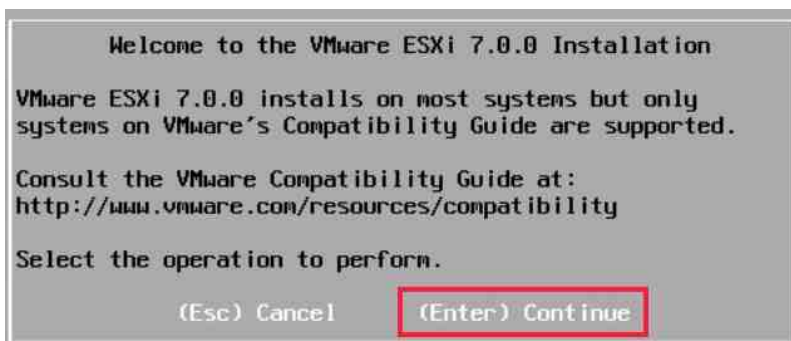
12. Right-click on sa-esxi-04-vm from the navigator pane and select Power > Power On.



13. Right-click on sa-esxi-04-vm once more and select Open Remote Console.



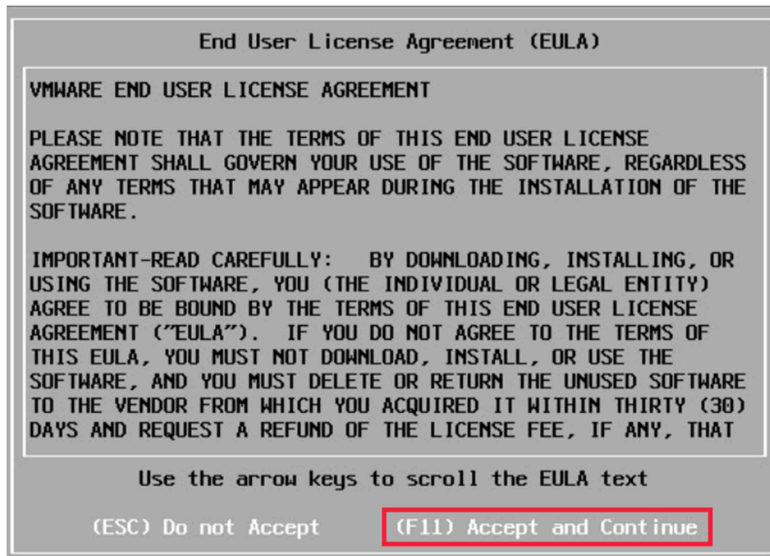
14. Notice a VMware Remote Console window appears, which provides direct console access to the machine. Once the ESXi installer initializes, notice the Welcome to the VMware ESXi 7.0.0 Installation message appears. To proceed with the installation, press the Enter key.



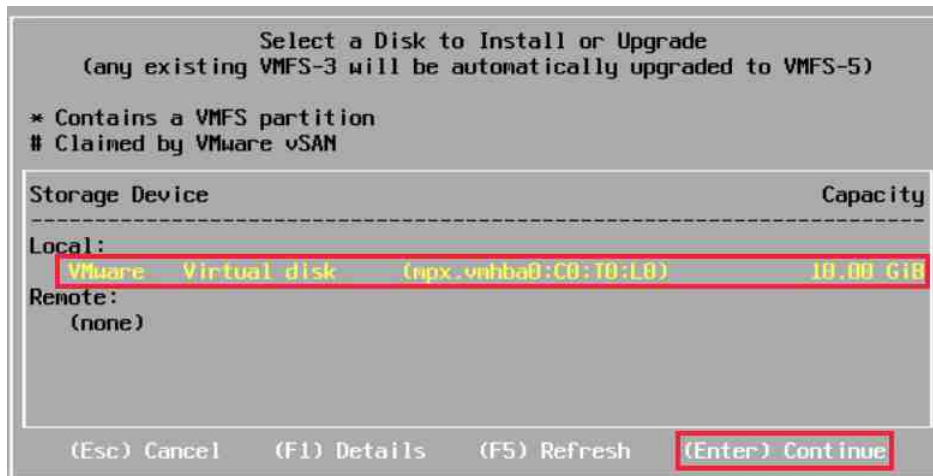


Due to VMware Tools not being installed for the machine, press CTRL+ALT to release the mouse pointer from the console when desired.

15. On the Accept the End-User License Agreement (EULA) step, press F11 to accept and continue.



16. On the Select a Disk to Install or Upgrade step, ensure that the local VMware virtual disk is selected and press the Enter key to continue.



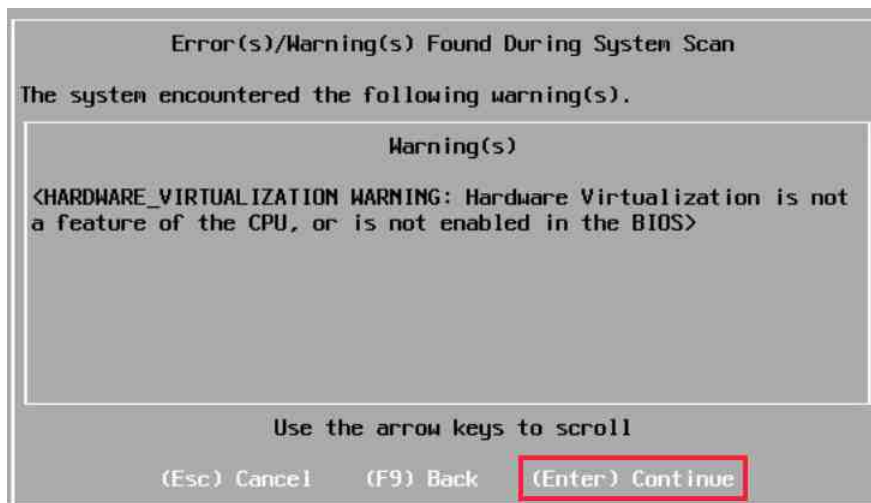
17. On the Please select a keyboard layout step, select US Default and press the Enter key.



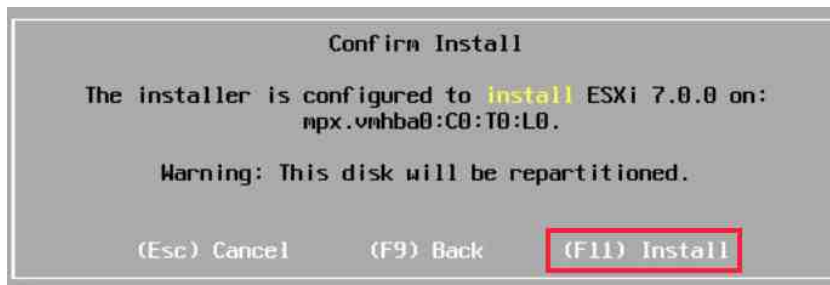
18. On the Enter a root password step, type Training\$ in the Root password and Confirm password fields. Press the Enter key to continue.



19. If an error appears regarding hardware virtualization, press the Enter key to continue.



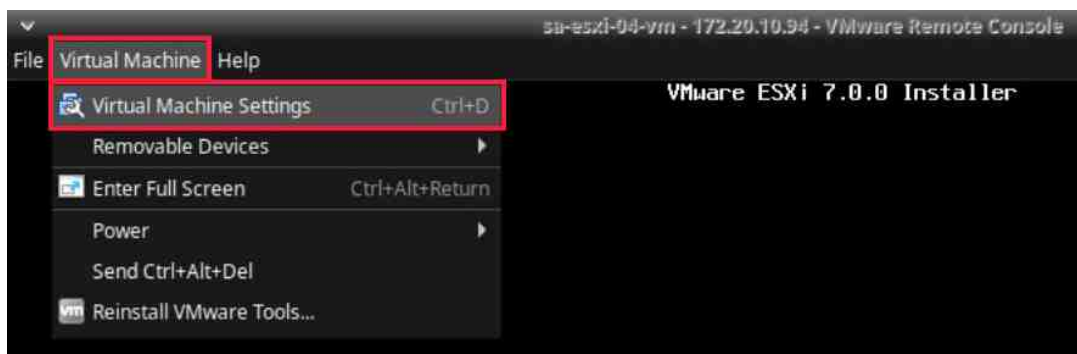
20. On the Confirm Install step, press F11 to confirm the installation.



**Please
Note**

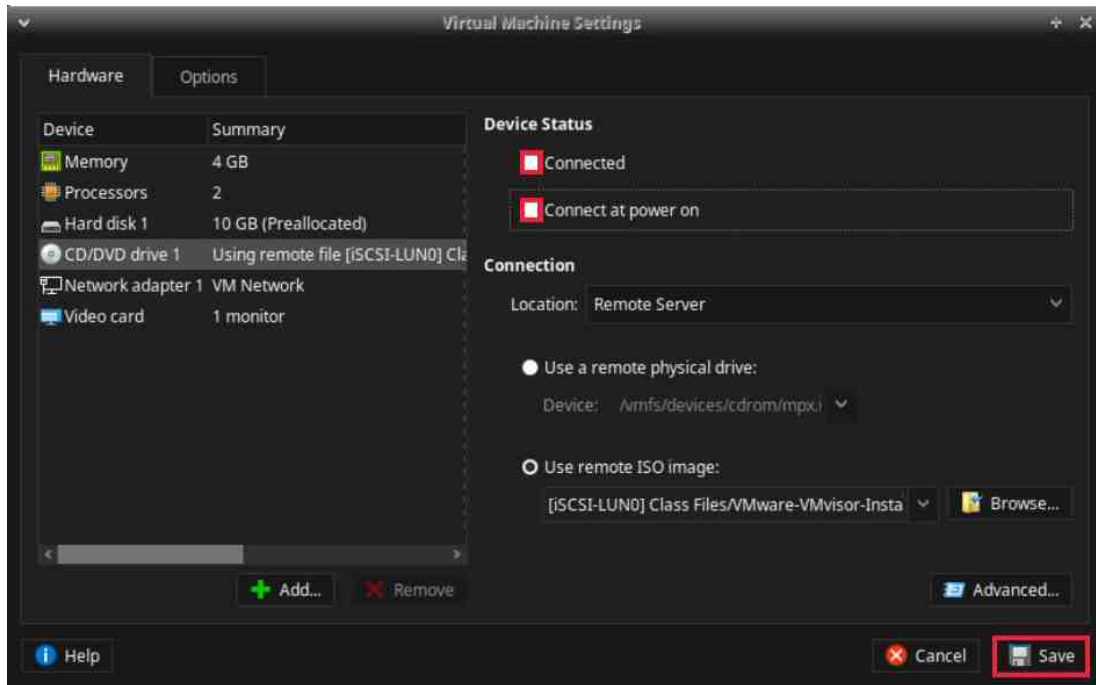
The installation may take about 3-5 minutes to complete. A progress bar shows the completion status of the installation.

21. When the Installation Complete message appears, disconnect the media by clicking on Virtual Machine > Virtual Machine Settings or by pressing CTRL+D.



Remember to press CTRL+ALT at the same time to release the mouse pointer from the virtual machine's console window.

22. In the Virtual Machine Settings window, click on CD/DVD drive 1 under the Hardware tab and disconnect the media by unchecking the Connected and Connect at power on checkboxes. Click Save.



23. Back on the Installation Complete message, press the Enter key to reboot the system

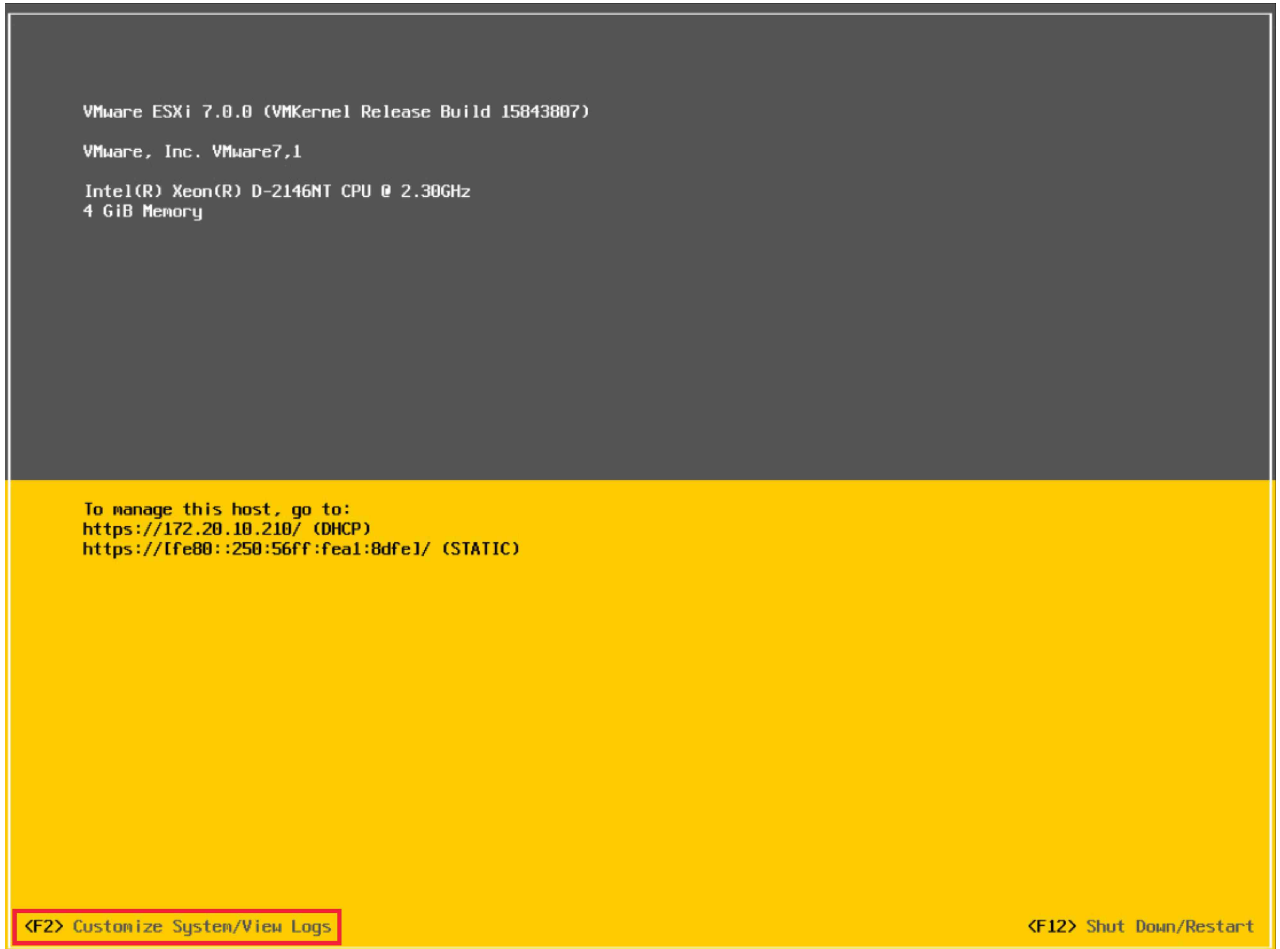


24. Monitor the reboot process until the Direct Console User Interface (DCUI) reappears. You have now successfully installed the VMware ESXi operating system. Leave the VMware Remote Console open and continue to the next task.

2 Configure an ESXi host

In this task, you will connect to the host Direct Console User Interface (DCUI) to view and modify the nested ESXi host configuration.

1. In the VMware Remote Console window, press F2 to customize the system or view the logs.



Due to VMware Tools not being installed for the machine, press CTRL+ALT to release the pointer from the console when desired.

2. Notice the Authentication Required dialog box appears; enter `root` as the Login Name and `Train1ng$` as the Password. Press Enter.



Authentication Required

Enter an authorized login name and password for localhost.netlab.local.

Configured Keyboard (US Default)

Login Name: [root]

Password: [*****_]

<Enter> OK <Esc> Cancel

3. On the System Customization screen, ensure that the Configure Password option is selected, and press Enter to change the nested host password.



System Customization

Configure Password

Configure Lockdown Mode

Configure Management Network

Restart Management Network

Test Management Network

Network Restore Options

Configure Keyboard

Troubleshooting Options

View System Logs

View Support Information

Reset System Configuration

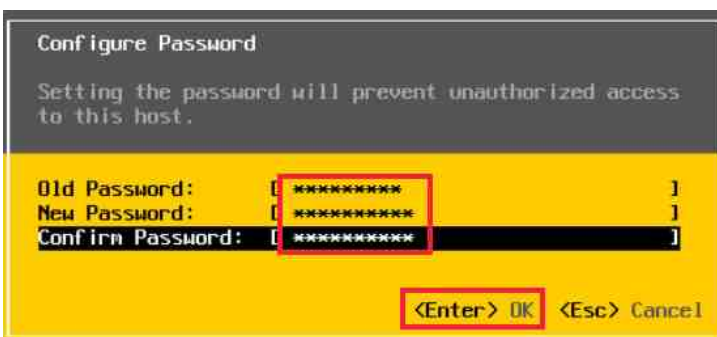
Configure Password

Set

To prevent unauthorized access to this system, set the password for the user.

<Enter> Change <Esc> Log Out

4. In the Configure Password dialog box, enter `Train1ng$` as the old password and type `vmware123!` into both the New Password and Confirm Password text fields. Press Enter.



Configure Password

Setting the password will prevent unauthorized access to this host.

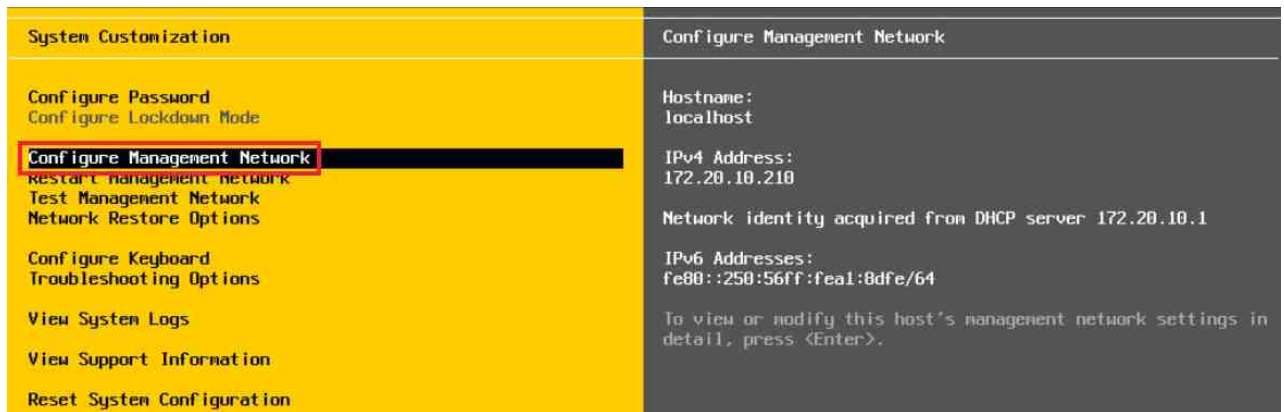
Old Password: [*****]

New Password: [*****]

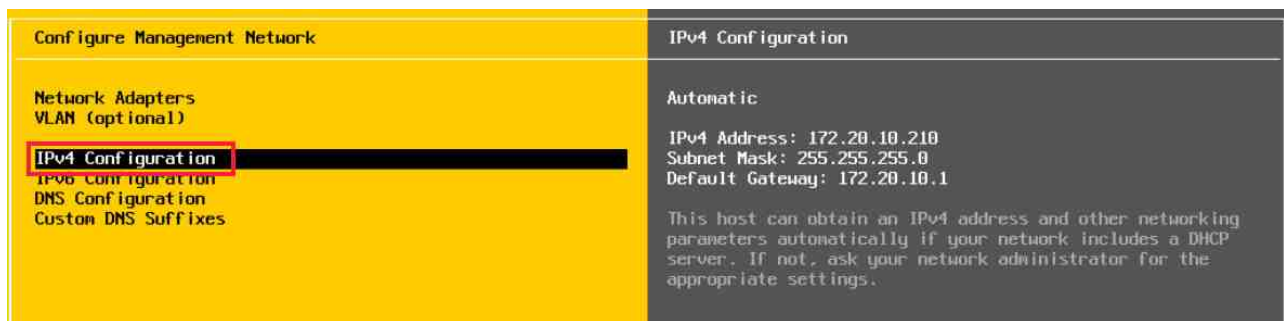
Confirm Password: [*****]

<Enter> OK <Esc> Cancel

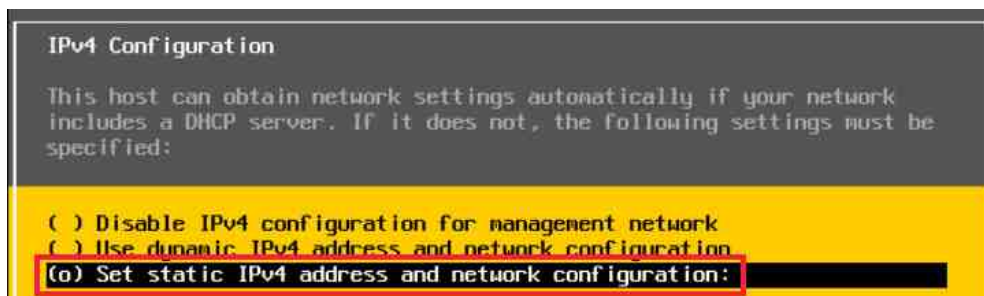
5. Select Configure Management Network. Press Enter.



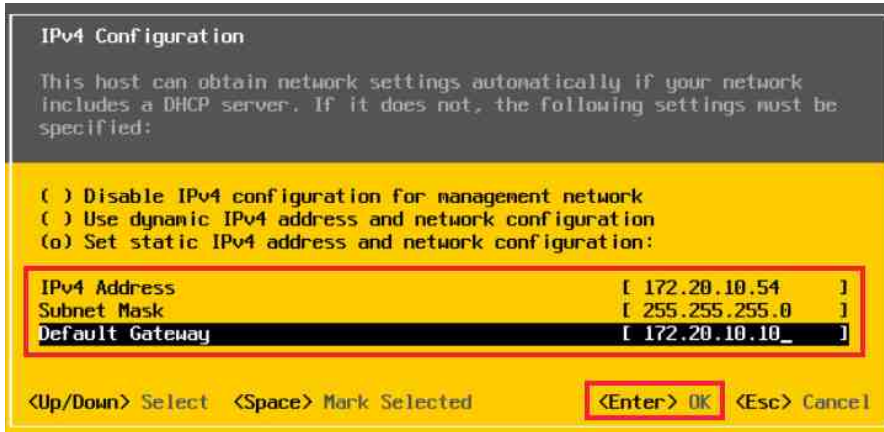
6. On the Configure Management Network screen, press the down arrow to select IPv4 Configuration. Press Enter.



7. In the IPv4 Configuration dialog box, go to Set static IPv4 address and network configuration and press the space bar to select it.



8. Press the down arrow to IPv4 Address and type in 172.20.10.54 followed by typing 255.255.255.0 as the Subnet Mask, and 172.20.10.10 as the Default Gateway. Press Enter to save the configurations.



IPv4 Configuration

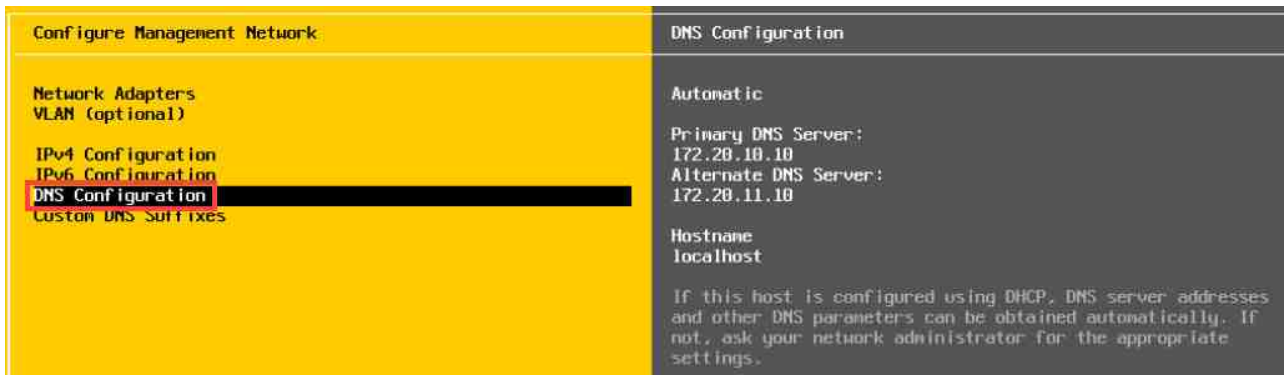
This host can obtain network settings automatically if your network includes a DHCP server. If it does not, the following settings must be specified:

- () Disable IPv4 configuration for management network
- () Use dynamic IPv4 address and network configuration
- (o) Set static IPv4 address and network configuration:

IPv4 Address	[172.20.10.54]
Subnet Mask	[255.255.255.0]
Default Gateway	[172.20.10.10]

<Up/Down> Select <Space> Mark Selected **<Enter> OK** <Esc> Cancel

9. On the Configure Management Network screen, use the down arrow to navigate to DNS Configuration and press Enter.



Configure Management Network

Network Adapters
VLAN (optional)

IPv4 Configuration
IPv6 Configuration
DNS Configuration
Custom DNS Suffixes

DNS Configuration

Automatic

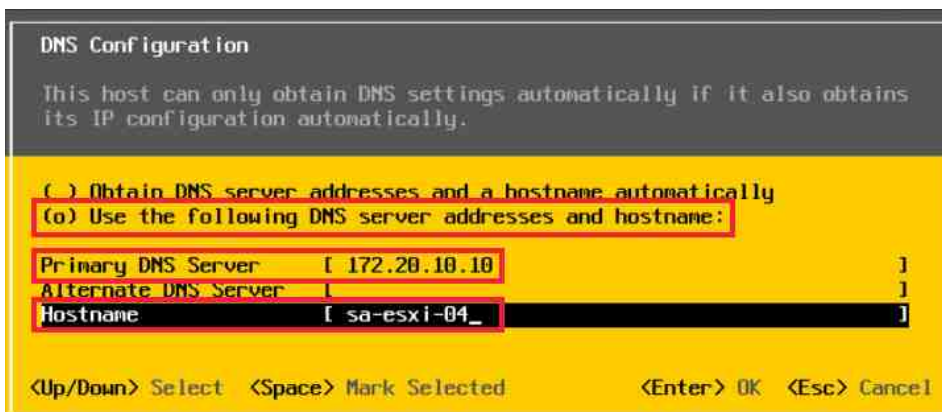
Primary DNS Server:
172.20.10.10

Alternate DNS Server:
172.20.11.10

Hostname
localhost

If this host is configured using DHCP, DNS server addresses and other DNS parameters can be obtained automatically. If not, ask your network administrator for the appropriate settings.

10. In the DNS Configuration dialog box, go to Use the following DNS server addresses and hostname and press the space bar to select it. Verify that 172.20.10.10 has been populated in the Primary DNS Server field; remove 172.20.10.11 in the Alternate DNS Server window. Type sa-esxi-04 into the Hostname field. Press Enter.



DNS Configuration

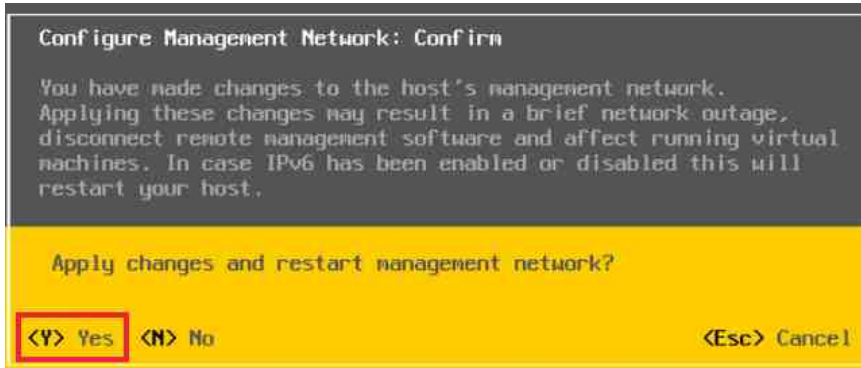
This host can only obtain DNS settings automatically if it also obtains its IP configuration automatically.

- () Obtain DNS server addresses and a hostname automatically
- (o) Use the following DNS server addresses and hostname:

Primary DNS Server	[172.20.10.10]
Alternate DNS Server	[]
Hostname	[sa-esxi-04]

<Up/Down> Select <Space> Mark Selected **<Enter> OK** <Esc> Cancel

11. Back on the Configure Management Network screen, press Esc to exit.
12. When prompted by the Configure Management Network: Confirm message, press Y to accept the changes.



13. Leave the ESXi DCUI opened to continue with the next task.

3 Enable ESXi Shell and SSH

In this task, you will configure the VMware ESXi host to enable ESXi Shell and SSH. With this enabled, you can run commands or scripts on your ESXi host in a secure fashion. By default, the ESXi Shell and SSH service are disabled. You will enable ESXi Shell to troubleshoot issues that may arise locally. An open SSH port may pose a potential security risk. You should enable it only as needed and disable it after the required task is completed.

1. While in the sa-esxi-04 DCUI, on the System Customization screen, select Troubleshooting Options. Press Enter.



2. On the Troubleshooting Options screen, use the down arrow to select Enable ESXi Shell and press Enter to enable the ESXi Shell.



Notice the selection option changes from Enable ESXi Shell to Disable ESXi Shell. In the right pane, you should see ESXi Shell is Enabled.

3. On the Troubleshooting Options screen, use the down arrow to select Enable SSH and press Enter to enable SSH.





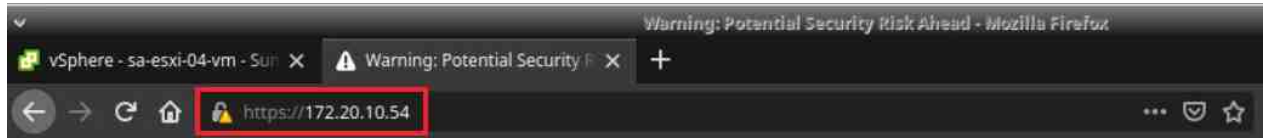
Notice the option changes from Enable SSH to Disable SSH. In the right pane, you should see SSH is Enabled.

4. Press Esc to return to the main options screen. You may close the sa-esxi-04-vm VMware Remote Console window and continue with the next task.

4 Add an ESXi Host to Active Directory

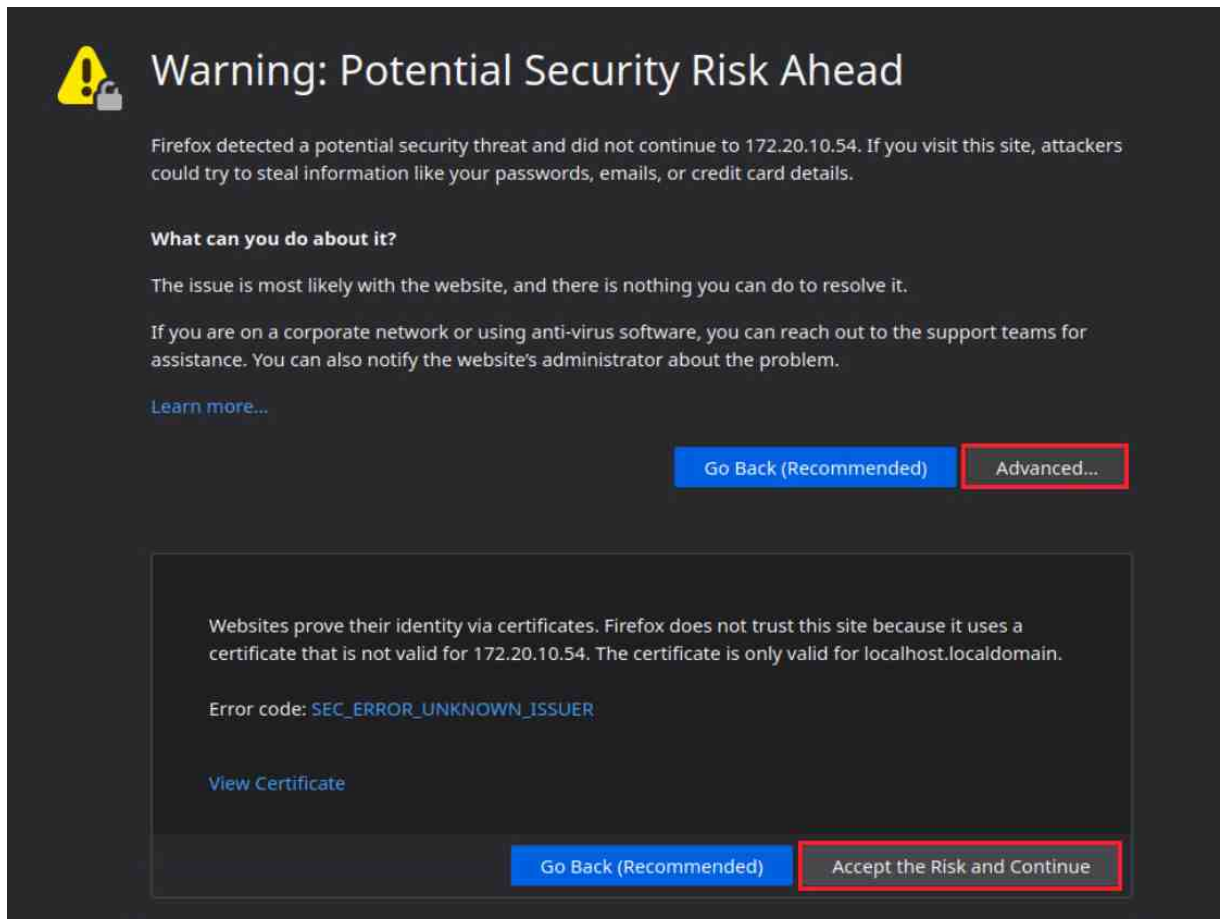
In this task, you will add an ESXi Host to Active Directory by accessing the ESXi Web Client.

1. Change focus to the Firefox Web Browser. Open a new tab, then type in `https://172.20.10.54` into the address field and press Enter.

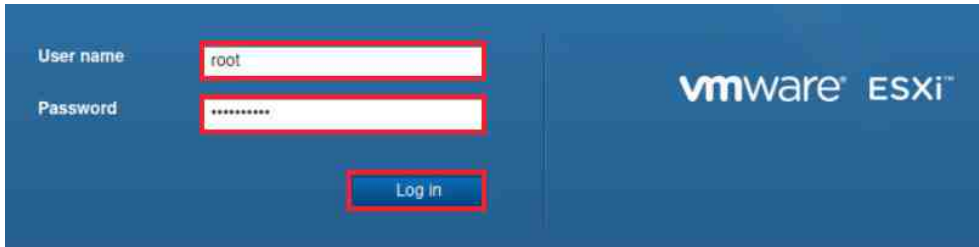


Note the IP address of 172.20.10.54 is the Management IP address that was set up in Task 2 of configuring sa-esxi-04. This is how you will access the ESXi Web Client to join sa-esxi-04 to an Active Directory domain. For this lab, Active Directory has been pre-configured.

2. On the Warning: Potential Security Risk Ahead page, click Advanced, and Accept the Risk and Continue.



- On the VMware ESXi Web Client page, type `root` as the Username and `vmware123!` as the Password. Click Log in.

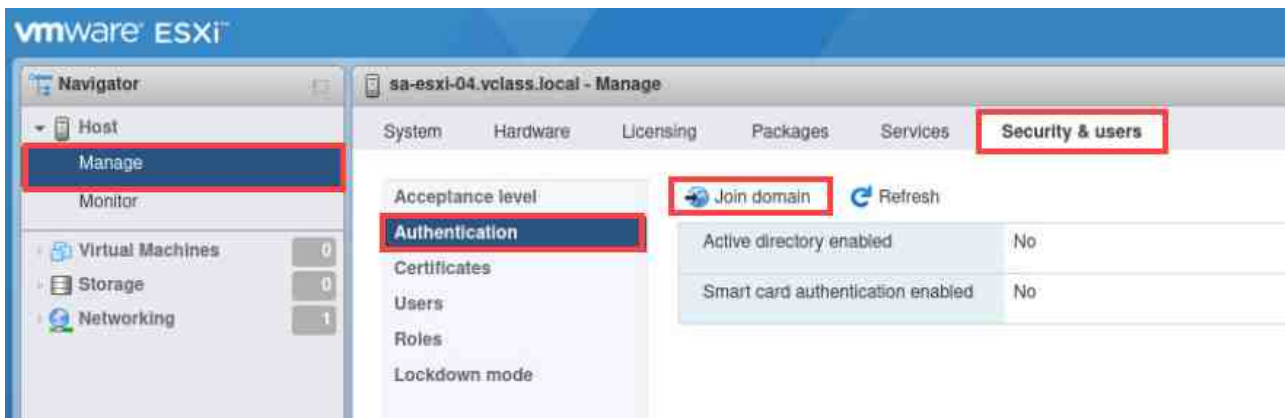


The image shows the VMware ESXi login page. It has a blue background with the VMware ESXi logo on the right. On the left, there are two input fields: 'User name' with 'root' entered and 'Password' with 'vmware123!' entered (masked with dots). Below these fields is a 'Log in' button. Red boxes highlight the input fields and the 'Log in' button.

Please Note

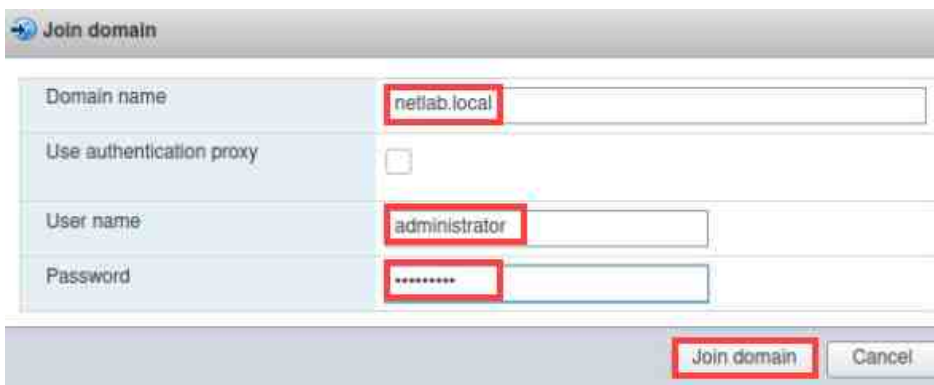
You may ignore the “Help us improve the VMware Host Client” message presented on the VMware ESXi Sign-On page by clicking OK and continue moving forward with the lab.

- In the Navigator pane, click on the Manage tab. In the `sa-esxi-04.vclass.local` – Manage window, click on Security & users, Authentication, and Join domain.



The image shows the VMware ESXi Manage console for the host `sa-esxi-04.vclass.local`. The 'Navigator' pane on the left has the 'Manage' tab selected. The main pane shows the 'Security & users' section, with 'Authentication' selected. The 'Join domain' button is highlighted. Below it, there are two rows of settings: 'Active directory enabled' and 'Smart card authentication enabled', both set to 'No'. Red boxes highlight the 'Manage' tab, 'Security & users', 'Authentication', and 'Join domain' button.

- In the Join Domain window, type `netlab.local` for the Domain name, type `administrator` for the Username, type `Training$` for the Password, and lastly, click Join Domain.

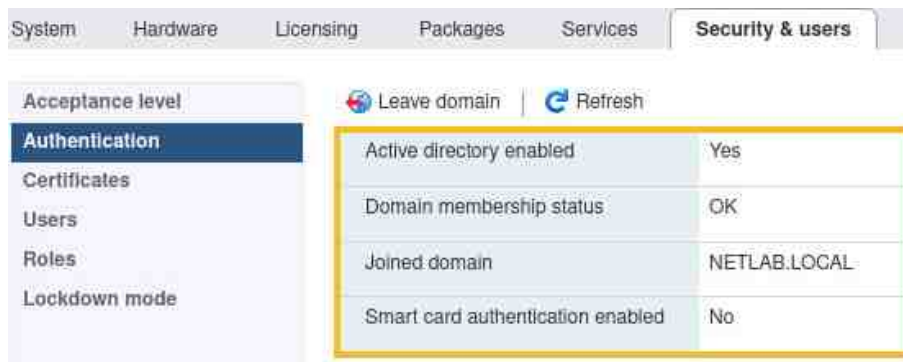


The image shows the 'Join domain' dialog box. It has four input fields: 'Domain name' with 'netlab.local' entered, 'Use authentication proxy' (unchecked), 'User name' with 'administrator' entered, and 'Password' with 'Training\$' entered (masked with dots). At the bottom right, there are 'Join domain' and 'Cancel' buttons. Red boxes highlight the 'Domain name', 'User name', 'Password', and 'Join domain' button.



For this lab, the user administrator has already been added to Active Directory and is a member of the ESX Admins domain group. This group has full administrative rights to join ESXi hosts to domains.

- On the sa-esxi-04.netlab.local web client window, you will notice that Active Directory enabled has a status of Yes. This means that you have successfully enabled Active Directory on the sa-esxi-04 host. Domain membership status shows OK and means that the domain controller has verified the sa-esxi-04 status as joined. Joined domain shows you have joined the NETLAB.LOCAL domain. Leave the ESXi Web Client open and continue to the next task.

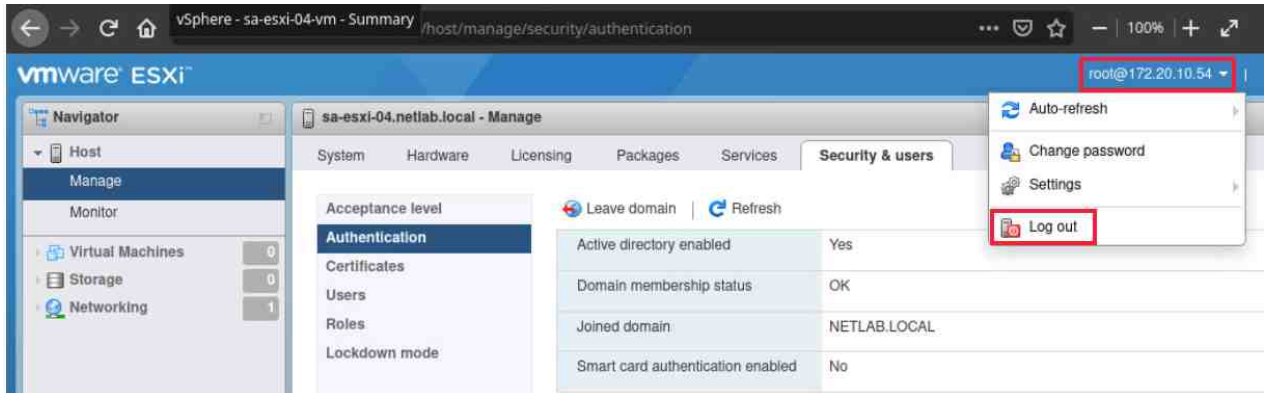


- Leave the VMware ESXi Client open and continue to the next task.

5 Log In to an ESXi host as an Active Directory User

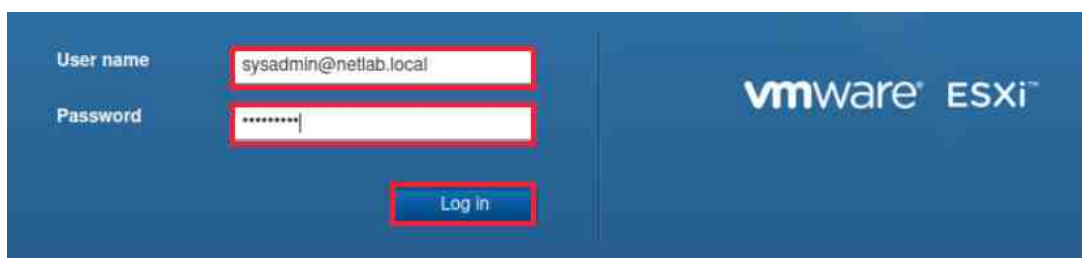
In this task, you will log in to sa-esxi-04 as the Active Directory user to verify the user `sysadmin@netlab.local` is a member of the ESXi administrator's domain group. This group is what allows Active Directory users to log in to an ESXi host that has been joined to a domain.

1. On the sa-esxi-04 Web Client page, log out of the root user by clicking on `root@172.20.10.54` and select Log out.

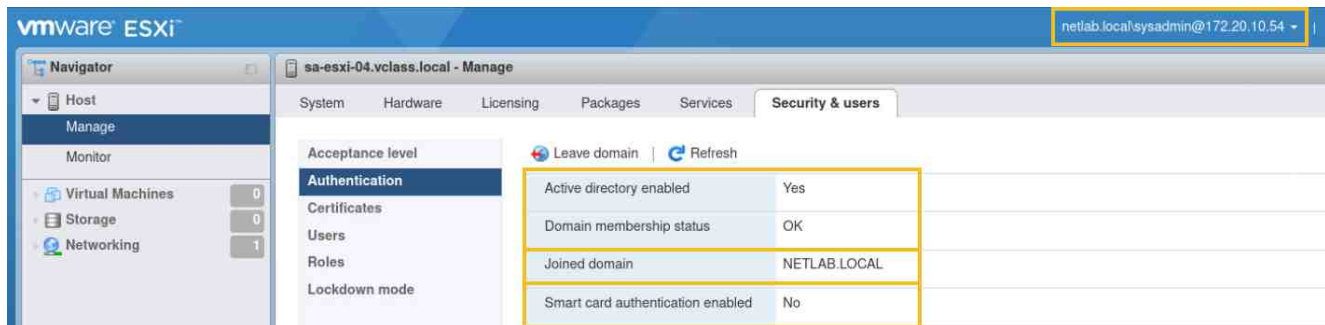


For this lab, the user `sysadmin@netlab.local` has already been added to Active Directory and is a member of the ESX Admins domain group. This group has full administrative rights to join ESXi hosts to domains.

2. On the VMWARE ESXi Web Client page, type `sysadmin@netlab.local` as the Username and `vmware123` as the Password. Click Log in.



3. In the navigator pane, click the Manage tab. Select Security & users and click Authentication. Verify once again that sa-esxi-04 is connected to the netlab.local domain. Notice the login was successful with the Active Directory user of netlab.local\sysadmin@172.20.10.54.



4. The lab is now complete; you may end the reservation.